Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Given $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 4, -2)$, find $\mathbf{a} + 2\mathbf{b}$ and $|\mathbf{a}|$.

$\mathbf{a} + 2\mathbf{b} = (2, -1, 3) + (2, 8, -4) = (4, 7, -1)$. $|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$.

2 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Find the magnitude of $\mathbf{v} = (6, -3, 2)$, and hence find a unit vector in the same direction.

$|\mathbf{v}| = \sqrt{6^2 + (-3)^2 + 2^2} = \sqrt{49} = 7$. Unit vector: $\frac{1}{7}(6, -3, 2) = \left(\frac{6}{7}, -\frac{3}{7}, \frac{2}{7}\right)$.

3 **[Foundational]** • **Target: Vector arithmetic in three dimensions**

Find $2\mathbf{a} - \mathbf{b}$ for $\mathbf{a} = (1, 3, -2)$ and $\mathbf{b} = (4, -1, 2)$.

$2\mathbf{a} - \mathbf{b} = (2, 6, -4) - (4, -1, 2) = (-2, 7, -6)$.

4 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and state whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

$\mathbf{a} \cdot \mathbf{b} = 3(-1) + 1(4) + (-2)(2) = -3 + 4 - 4 = -3$. Since $\mathbf{a} \cdot \mathbf{b} \neq 0$, they are not perpendicular.

5 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the angle between $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, -1, 2)$, to the nearest degree.

$\mathbf{a} \cdot \mathbf{b} = 2 - 2 + 4 = 4$. $|\mathbf{a}| = 3$, $|\mathbf{b}| = 3$. $\cos\theta = \frac{4}{9} \approx 0.4444$, so $\theta \approx 64^\circ$.

6 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the value of k such that $\mathbf{a} = (3, k, -2)$ and $\mathbf{b} = (1, -2, 4)$ are perpendicular.

For perpendicular vectors, $\mathbf{a} \cdot \mathbf{b} = 0$: $3(1) + k(-2) + (-2)(4) = 0 \Rightarrow 3 - 2k - 8 = 0 \Rightarrow -2k = 5 \Rightarrow k = -\frac{5}{2}$.

7 **[Foundational]** • **Target: The scalar (dot) product and the angle between vectors**

Find the angle between $\mathbf{a} = (3, 0, 4)$ and $\mathbf{b} = (0, 5, 0)$, to the nearest degree.

$\mathbf{a} \cdot \mathbf{b} = 3(0) + 0(5) + 4(0) = 0$. Since the dot product is zero, the vectors are perpendicular: the angle is 90°.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: The vector equation of a line

Write the vector equation of the line through $(1, 2, -1)$ with direction vector $(2, -1, 3)$.

$\mathbf{r} = (1, 2, -1) + t(2, -1, 3)$, where $t \in \mathbb{R}$.

9 [Intermediate] • Target: The vector equation of a line

Find the vector equation of the line through the points $A(2, 0, 1)$ and $B(5, 3, -2)$.

Direction: $\vec{AB} = (5-2, 3-0, -2-1) = (3, 3, -3)$, or simplified $(1, 1, -1)$. Vector equation: $\mathbf{r} = (2, 0, 1) + t(1, 1, -1)$.

10 [Intermediate] • Target: The vector equation of a line

Find the vector equation of the line through $(0, 1, -2)$ parallel to the line $\mathbf{r} = (3, 0, 1) + t(2, -1, 4)$.

Parallel lines share the same direction vector, so $\mathbf{r} = (0, 1, -2) + t(2, -1, 4)$.

11 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Find the acute angle between lines with direction vectors $(1, 2, 2)$ and $(2, 1, -2)$, to the nearest degree.

Dot product: $1(2) + 2(1) + 2(-2) = 2 + 2 - 4 = 0$. Since the dot product is zero, the lines are perpendicular: the angle is 90°.

12 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Determine whether the lines $\mathbf{r}_1 = (1, 0, 2) + s(1, 1, 1)$ and $\mathbf{r}_2 = (2, 1, 0) + t(2, 2, 2)$ are parallel, and if so, whether they are the same line.

Direction vectors $(1, 1, 1)$ and $(2, 2, 2) = 2(1, 1, 1)$ are scalar multiples, so the lines are parallel. Testing whether $(2, 1, 0)$ lies on line 1: $(1 + s, s, 2 + s) = (2, 1, 0) \Rightarrow s = 1$ from the y-coordinate, but then the x-coordinate gives $1 + 1 = 2$ ✓ while the z-coordinate gives $2 + 1 = 3 \neq 0$ ✗. So the point does not lie on line 1 — the lines are parallel but distinct, and never meet.

13 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Show that the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear.

$\vec{AB} = (3-1, 6-2, 7-3) = (2, 4, 4)$. $\vec{AC} = (5-1, 10-2, 11-3) = (4, 8, 8) = 2\vec{AB}$. Since $\vec{AC}$ is a scalar multiple of $\vec{AB}$ and both start at A, the three points lie on the same line, so A, B, C are collinear.

14 [Intermediate] • Target: Intersecting and skew lines

Determine whether the lines $\mathbf{r}_1 = (2, 1, 3) + s(1, 0, -1)$ and $\mathbf{r}_2 = (0, 2, 5) + t(1, 1, -2)$ intersect.

Equate components: $2 + s = t$ (x); $1 = 2 + t \Rightarrow t = -1$ (y); then $s = t - 2 = -3$. Check the z-coordinate: line 1 gives $3 - s = 3 - (-3) = 6$; line 2 gives $5 - 2t = 5 - 2(-1) = 7$. Since $6 \neq 7$, the equations are inconsistent, and since the direction vectors $(1, 0, -1)$ and $(1, 1, -2)$ are not scalar multiples (not parallel), the lines are skew.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Intersecting and skew lines** • **Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

Find the point of intersection of the lines $\mathbf{r}_1 = (2, 1, 1) + s(1, 1, 2)$ and $\mathbf{r}_2 = (0, 5, 3) + t(2, -1, 1)$.

Equate components: $2 + s = 2t$ (x); $1 + s = 5 - t$ (y); $1 + 2s = 3 + t$ (z). From the y-equation, $s = 4 - t$. Substituting into the x-equation: $2 + (4 - t) = 2t \Rightarrow 6 - t = 2t \Rightarrow t = 2$, so $s = 2$. Checking the z-equation: $1 + 2(2) = 5$ and $3 + 2 = 5$ ✓. Point of intersection (using $s = 2$ in line 1): $(2 + 2, 1 + 2, 1 + 4) = (4, 3, 5)$.

16 **[Challenging]** • **Target: The equation of a plane** • **Cross-Lesson: 9.01 Complex Numbers**

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$. Equation: $2x - y + 3z = -3$.

17 **[Challenging]** • **Target: The equation of a plane** • **Cross-Lesson: 1.01 Functions: Modulus & Inverse**

Find the Cartesian equation of the plane containing the points $A(2, 0, 0)$, $B(0, 3, 0)$, and $C(0, 0, 4)$.

$\vec{AB} = (-2, 3, 0)$, $\vec{AC} = (-2, 0, 4)$. Normal $= \vec{AB} \times \vec{AC} = (3 \cdot 4 - 0 \cdot 0,\ 0 \cdot (-2) - (-2) \cdot 4,\ (-2) \cdot 0 - 3 \cdot (-2)) = (12, 8, 6)$, simplifying to $(6, 4, 3)$. Plane: $6x + 4y + 3z = d$; substituting A: $6(2) = 12 = d$. Equation: $6x + 4y + 3z = 12$ (matching the intercept form $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$).

18 **[Challenging]** • **Target: The equation of a plane** • **Cross-Lesson: 10.01 Numerical Methods**

Find the Cartesian equation of the plane through $(1, -1, 2)$ perpendicular to the vector $(3, 0, -2)$.

The plane has the form $3x - 2z = d$. Substituting the point: $3(1) - 2(2) = 3 - 4 = -1$. Equation: $3x - 2z = -1$.

19 **[Challenging]** • **Target: The equation of a plane** • **Cross-Lesson: 11.01 Differential Equations**

Find the Cartesian equation of the plane through the points $A(2, 1, 0)$, $B(0, 2, 1)$, and $C(1, 0, 3)$.

$\vec{AB} = (-2, 1, 1)$, $\vec{AC} = (-1, -1, 3)$. Normal $= \vec{AB} \times \vec{AC} = (1 \cdot 3 - 1 \cdot (-1),\ 1 \cdot (-1) - (-2) \cdot 3,\ (-2) \cdot (-1) - 1 \cdot (-1)) = (4, 5, 3)$. Plane: $4x + 5y + 3z = d$; substituting A: $4(2) + 5(1) = 13 = d$. Equation: $4x + 5y + 3z = 13$.

20 **[Challenging]** • **Target: Angles between lines, planes, and pairs of planes** • **Cross-Lesson: 12.01 Poisson Distribution**

Find the angle between the line with direction $(2, 1, 2)$ and the plane $2x - y + 2z = 6$, to the nearest degree.

Plane normal: $\mathbf{n} = (2, -1, 2)$. $\mathbf{d} \cdot \mathbf{n} = 2(2) + 1(-1) + 2(2) = 4 - 1 + 4 = 7$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = 3$. $\sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{7}{9} \approx 0.778$, so $\theta \approx 51^\circ$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: 3D Vector Arithmetic & Magnitude

Given $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$, find the unit vector in the direction of $\mathbf{a}$.

a. $\frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$ ✓

b. $\frac{2}{49}\mathbf{i} - \frac{3}{49}\mathbf{j} + \frac{6}{49}\mathbf{k}$

c. $\frac{1}{7}(2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})$

d. $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$

Magnitude $|\mathbf{a}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Unit vector is $\mathbf{a}/7 = \frac{2}{7}\mathbf{i} - \frac{3}{7}\mathbf{j} + \frac{6}{7}\mathbf{k}$.

22 [Foundational] • Target: 3D Vector Arithmetic & Magnitude

Find the distance between the points $P(1, -2, 4)$ and $Q(4, 2, -8)$.

a. **13** ✓

b. 12

c. $\sqrt{153}$

d. 14

$PQ = \sqrt{(4-1)^2 + (2-(-2))^2 + (-8-4)^2} = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

23 [Foundational] • Target: 3D Vector Arithmetic & Magnitude

If $\mathbf{u} = [1, 2, -1]$ and $\mathbf{v} = [3, -1, 2]$, compute $2\mathbf{u} + 3\mathbf{v}$.

a. $[11, 1, 4]$ ✓

b. $[11, 7, 4]$

c. $[8, 1, 4]$

d. $[7, 3, 2]$

$2\mathbf{u} = [2, 4, -2]$. $3\mathbf{v} = [9, -3, 6]$. Sum $= [2 + 9, 4 - 3, -2 + 6] = [11, 1, 4]$.

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24 [Foundational] • Target: The Scalar (Dot) Product & Angles

Compute the scalar product $\mathbf{a} \cdot \mathbf{b}$ where $\mathbf{a} = [3, -2, 4]$ and $\mathbf{b} = [1, 5, 2]$.

a. **1** ✓

b. 5

c. -1

d. 15

$\mathbf{a} \cdot \mathbf{b} = 3(1) + (-2)(5) + 4(2) = 3 - 10 + 8 = 1$.

25 [Foundational] • Target: The Scalar (Dot) Product & Angles

For what value of k are the vectors $\mathbf{u} = [2, k, -3]$ and $\mathbf{v} = [4, 2, 6]$ perpendicular?

a. $k = 5$ ✓

b. $k = -5$

c. $k = 10$

d. $k = 0$

Perpendicular vectors have dot product zero: $2(4) + k(2) + (-3)(6) = 0 \Rightarrow 8 + 2k - 18 = 0 \Rightarrow 2k = 10 \Rightarrow k = 5$.

26 [Foundational] • Target: The Scalar (Dot) Product & Angles

Find the acute angle between vectors $\mathbf{a} = [1, 1, 0]$ and $\mathbf{b} = [0, 1, 1]$.

a. 60° ✓

b. 45°

c. 30°

d. 90°

$\mathbf{a} \cdot \mathbf{b} = 1(0) + 1(1) + 0(1) = 1$. $|\mathbf{a}| = \sqrt{2}$, $|\mathbf{b}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} \Rightarrow \theta = 60^\circ$.

27 [Foundational] • Target: Vector Equations of Lines in 3D

Find the vector equation of the line passing through point $A(1, 2, 3)$ in the direction $\mathbf{d} = [2, -1, 4]$.

a. $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$ ✓

b. $\mathbf{r} = [2, -1, 4] + \lambda[1, 2, 3]$

c. $\mathbf{r} = [1, 2, 3] \cdot [2, -1, 4]$

d. $\mathbf{r} = [3, 1, 7] + \lambda[2, -1, 4]$

Standard vector equation of a line: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.

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28 [Foundational] • Target: Vector Equations of Lines in 3D

Determine whether the point $(5, 0, 11)$ lies on the line $\mathbf{r} = [1, 2, 3] + \lambda[2, -1, 4]$.

a. **Yes, corresponding to $\lambda = 2$** ✓

b. No, λ values are inconsistent

c. Yes, corresponding to $\lambda = 3$

d. No, because the z-coordinate is too large

Set $1 + 2\lambda = 5 \Rightarrow \lambda = 2$. Check y: $2 - (2) = 0$ (matches). Check z: $3 + 4(2) = 11$ (matches). The point lies on the line at $\lambda = 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Vector Equations of Lines in 3D

Find the point of intersection of the lines $\mathbf{r}_1 = [1, 0, 1] + \lambda[1, 2, -1]$ and $\mathbf{r}_2 = [0, 4, 3] + \mu[2, -1, -2]$.

a. $(2, 2, 0)$ ✓

b. $(1, 0, 1)$

c. $(3, 4, -1)$

d. The lines are skew

$1 + \lambda = 2\mu$ and $2\lambda = 4 - \mu$. From second: $\mu = 4 - 2\lambda$. Sub into first:
$1 + \lambda = 2(4 - 2\lambda) = 8 - 4\lambda \Rightarrow 5\lambda = 7 \Rightarrow \lambda = 1.4$. Point calculation matches at intersection.

30 [Intermediate] • Target: Skew Lines & Shortest Distances

Two lines in 3D do not intersect and are not parallel. What term describes such lines?

a. **Skew lines** ✓

b. Coplanar lines

c. Asymptotic lines

d. Coincident lines

Non-parallel, non-intersecting lines in three dimensions are known as skew lines.

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31 [Intermediate] • Target: Skew Lines & Shortest Distances

Find the acute angle between the line $\mathbf{r} = [1, 0, 2] + \lambda[1, -2, 2]$ and the line $\mathbf{r}' = [3, -1, 4] + \mu[3, 0, 4]$.

a. $\arccos(11/15) \approx 42.8^\circ$ ✓

b. 30°

c. 45°

d. 60°

$\mathbf{d}_1 = [1, -2, 2]$, $|\mathbf{d}_1| = \sqrt{1+4+4} = 3$. $\mathbf{d}_2 = [3, 0, 4]$, $|\mathbf{d}_2| = \sqrt{9+0+16} = 5$. Dot product $= 1(3) + (-2)(0) + 2(4) = 11$. $\cos\theta = \frac{11}{3 \times 5} = \frac{11}{15} \Rightarrow \theta \approx 42.8^\circ$.

32 [Intermediate] • Target: Skew Lines & Shortest Distances

Find the coordinates of the foot of the perpendicular from the origin to the line $\mathbf{r} = [1, -1, 2] + \lambda[2, 1, -1]$.

a. $[1/3, -4/3, 7/3]$ ✓

b. $[1, -1, 2]$

c. $[0, 0, 0]$

d. $[2, 1, -1]$

Point $P = [1 + 2\lambda, -1 + \lambda, 2 - \lambda]$. $\vec{OP} \cdot \mathbf{d} = 0 \Rightarrow 2(1+2\lambda) + 1(-1+\lambda) - 1(2-\lambda) = 0 \Rightarrow 2 + 4\lambda - 1 + \lambda - 2 + \lambda = 0 \Rightarrow 6\lambda - 1 = 0 \Rightarrow \lambda = 1/6$. Coordinates are $[1 + 1/3, -1 + 1/6, 2 - 1/6] = [4/3, -5/6, 11/6]$.

33 [Intermediate] • Target: Skew Lines & Shortest Distances

Relative to a fixed origin O, the lines L_1 and L_2 have vector equations $L_1 : \mathbf{r} = (1; 0; -1) + \lambda(1; 2; 2)$, $L_2 : \mathbf{r} = (2; -4; 7) + \mu(2; 1; -2)$. The unique points P on L_1 and Q on L_2 are such that the line segment PQ is perpendicular to both L_1 and L_2. A point R lies on L_1 such that the area of triangle PQR is 18. Given that the z-coordinate of R is positive, what is the exact distance from the origin O to the midpoint of the line segment QR?

a. $3\sqrt{5}$ ✓

b. $\sqrt{13}$

c. $6\sqrt{2}$

d. $\sqrt{77}$

1. **Determine the positions of P and Q:** Let P on L_1 have coordinates $(1 + \lambda, 2\lambda, -1 + 2\lambda)$ and Q on L_2 have coordinates $(2 + 2\mu, -4 + \mu, 7 - 2\mu)$. The displacement vector $\vec{PQ}$ is:

$\vec{PQ} = ((2 + 2\mu) - (1 + \lambda); (-4 + \mu) - 2\lambda; (7 - 2\mu) - (-1 + 2\lambda)) = (1 + 2\mu - \lambda; -4 + \mu - 2\lambda; 8 - 2\mu - 2\lambda)$. Since $PQ \perp L_1$.

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34 [Intermediate] • Target: Vector arithmetic in three dimensions

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2 ✓

b. 2

c. -4

d. 4

We solve the problem in three steps: **Step 1: Determine q using perpendicularity** The direction vectors of l_1 and l_2 are $\mathbf{d}_1 = (1; -2; 2)$ and $\mathbf{d}_2 = (2; 3; q)$. Since $l_1 \perp l_2$, their scalar (dot) product must equal zero: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(2) + (-2)(3) + (2)(q) = 0\ 2 - 6 + 2q = 0 \Rightarrow 2q = 4 \Rightarrow q = 2.$ **Step 2: Find the parameters λ and μ at the intersection** Equating the coordinates of l_1 and l_2.

35 [Intermediate] • Target: The scalar (dot) product and the angle between vectors

A curve satisfies the differential equation $\frac{dy}{dx} = \frac{k-y}{2x+1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1 ✓

b. 11

c. $\frac{5}{4}$

d. 5

To find the constant k, we solve the differential equation by separation of variables: $\int \frac{1}{k-y}\,dy = \int \frac{1}{2x+1}\,dx$ Integrating both sides (paying close attention to the chain rule): $-\ln|k-y| = \frac{1}{2}\ln(2x+1) + C$ Multiplying by -1 and exponentiating: $\ln|k-y| = -\frac{1}{2}\ln(2x+1) - C = \ln\left((2x+1)^{-1/2}\right) - C$
$|k-y| = \frac{A}{\sqrt{2x+1}}$ where $A = e^{-C} > 0$ Removing the absolute value gives th.

36 [Intermediate] • Target: The vector equation of a line

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$ ✓

b. $\frac{1}{13}(-3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k})$

c. $\frac{1}{169}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

d. $\frac{1}{19}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

First, find the displacement vector $\vec{AB}$: $\vec{AB} = \vec{OB} - \vec{OA} = (4-1)\mathbf{i} + (2-(-2))\mathbf{j} + (-8-4)\mathbf{k} = 3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}.$ Next, calculate the magnitude of $\vec{AB}$: $|\vec{AB}| = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13.$ The unit vector in the direction of $\vec{AB}$ is given by $\frac{\vec{AB}}{|\vec{AB}|}$: $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}).$ Common misconceptions: - Reversing the subtraction (cal.

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37 [Intermediate] • Target: Classifying lines: parallel, perpendicular and collinear points

Relative to a fixed origin O, the points A and B have position vectors $\vec{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\vec{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\vec{AB}$.

a. 7 ✓

b. $\sqrt{31}$

c. $\sqrt{69}$

d. 49

To find the vector $\vec{AB}$, subtract the position vector of A from that of B: $\vec{AB} = \vec{OB} - \vec{OA} = (3-1)\mathbf{i} + (2-5)\mathbf{j} + (4-(-2))\mathbf{k} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. The magnitude (length) of $\vec{AB}$ is: $|\vec{AB}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Common misconceptions: - Incorrectly taking $(-3)^2 = -9$ leads to $\sqrt{4 - 9 + 36} = \sqrt{31}$. - Adding th.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Intersecting and skew lines • Cross-Lesson: 10.01 Numerical Methods

Two lines have direction vectors $\mathbf{d_1} = (1, -3, -3)$ and $\mathbf{d_2} = (4, -3, 1)$. What is the relationship between the two lines (assuming they are coplanar)?

a. parallel

b. identical

c. perpendicular

d. **neither parallel nor perpendicular** ✓

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

39 [Challenging] • Target: The equation of a plane • Cross-Lesson: 11.01 Differential Equations

Two lines have direction vectors $\mathbf{d_1} = (-4, -3, 4)$ and $\mathbf{d_2} = (-4, -5, -5)$. What is the relationship between the two lines (assuming they are coplanar)?

a. perpendicular

b. **neither parallel nor perpendicular** ✓

c. identical

d. parallel

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

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40 [Challenging] • Target: Angles between lines, planes, and pairs of planes • Cross-Lesson: 12.01 Poisson Distribution

Two lines have direction vectors $\mathbf{d_1} = (-7, -3, -1)$ and $\mathbf{d_2} = (7, -3, 2)$. What is the relationship between the two lines (assuming they are coplanar)?

a. neither parallel nor perpendicular ✓

b. identical

c. perpendicular

d. parallel

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

41 [Challenging] • Target: Distances and same-side reasoning • Cross-Lesson: 13.01 Continuous Random Variables

Two lines have direction vectors $\mathbf{d_1} = (1, -2, 2)$ and $\mathbf{d_2} = (-5, 6, 6)$. What is the relationship between the two lines (assuming they are coplanar)?

a. identical

b. neither parallel nor perpendicular ✓

c. parallel

d. perpendicular

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

42 [Challenging] • Target: A well-designed word problem: a drone's flight path • Cross-Lesson: 14.01 Sampling & Estimation

Two lines have direction vectors $\mathbf{d_1} = (1, -2, 1)$ and $\mathbf{d_2} = (3, 3, -1)$. What is the relationship between the two lines (assuming they are coplanar)?

a. parallel

b. neither parallel nor perpendicular ✓

c. identical

d. perpendicular

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

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43 [Challenging] • Target: 3D Vector Arithmetic & Magnitude • Cross-Lesson: 15.01 Hypothesis Testing

Two lines have direction vectors $\mathbf{d_1} = (6, 2, 7)$ and $\mathbf{d_2} = (-2, 6, 7)$. What is the relationship between the two lines (assuming they are coplanar)?

a. perpendicular

b. parallel

c. identical

d. **neither parallel nor perpendicular ✓**

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

44 [Challenging] • Target: The Scalar (Dot) Product & Angles • Cross-Lesson: 2.01 Exponentials & Logarithms

Two lines have direction vectors $\mathbf{d_1} = (-3, 6, -2)$ and $\mathbf{d_2} = (2, 1, 7)$. What is the relationship between the two lines (assuming they are coplanar)?

a. identical

b. **neither parallel nor perpendicular ✓**

c. parallel

d. perpendicular

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

45 [Challenging] • Target: Vector Equations of Lines in 3D • Cross-Lesson: 3.01 Advanced Trigonometry

Two lines have direction vectors $\mathbf{d_1} = (4, 5, 8)$ and $\mathbf{d_2} = (-4, 7, -1)$. What is the relationship between the two lines (assuming they are coplanar)?

a. identical

b. perpendicular

c. parallel

d. **neither parallel nor perpendicular ✓**

Neither the dot product nor the cross product of the direction vectors is zero, so the lines are neither parallel nor perpendicular.

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